

Support Vector Machines

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Hard-Margin SVM

Setting

- Given (x_i, y_i) pairs, $x_i \in \mathbf{R}^d$, $y_i \in \{-1, +1\}$
- Dataset is linearly separable
 - (for now, while we're considering *hard-margin SVMs*)
- (Draw perceptron, highlight non-uniqueness drawback)
- Support Vector Machines (SVMs) try to find the “best” solution

Margins

- Suppose we have a perceptron solution w', b'
- If $\|w'\|_2 = 1$, then distance from point i to hyperplane is
$$\gamma_i = y_i(\langle w', x_i \rangle + b')$$
- (Draw picture)
- Margin of a dataset (wrt solution w', b') is $\gamma = \min_i \gamma_i$
- Perceptron tries to find any w', b' such that $\min_i \gamma_i > 0$
- SVM: Like perceptron, but try to maximize margin

Deriving SVM problem

- SVM: Like perceptron, but try to maximize margin

$$\max_{w', b'} \gamma, \text{ s.t. } \|w'\|_2 = 1, y_i(\langle w', x_i \rangle + b') \geq \gamma \text{ for all } i$$

- Substitute $w' = \gamma w, b' = \gamma b$

$$\max_{\gamma w, \gamma b} \gamma, \text{ s.t. } \|w\|_2 = 1/\gamma, y_i(\langle \gamma w, x_i \rangle + \gamma b) \geq \gamma \text{ for all } i$$

$$\max_{\gamma w, \gamma b} \gamma, \text{ s.t. } \|w\|_2 = 1/\gamma, y_i(\langle w, x_i \rangle + b) \geq 1 \text{ for all } i$$

$$\max_{\gamma w, \gamma b} \frac{1}{\|w\|_2}, \text{ s.t. } y_i(\langle w, x_i \rangle + b) \geq 1 \text{ for all } i$$

$$\min_{w, b} \frac{1}{2} \|w\|_2^2 \text{ s.t. } y_i(\langle w, x_i \rangle + b) \geq 1 \text{ for all } i$$

Hard-Margin SVM problem

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 \text{ s. t. } y_i(\langle w, x_i \rangle + b) \geq 1 \text{ for all } i$$

- Instead of keeping $\|w\|_2$ fixed and maximizing γ , do opposite
- Let $\hat{y}_i = \langle w, x_i \rangle + b$. Note that sign of $\hat{y}_i$ gives prediction

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 \text{ s. t. } y_i \hat{y}_i \geq 1 \text{ for all } i$$

- Compare with (weird writing of) perceptron's objective

$$\min_{w,b} 0 \text{ s. t. } y_i \hat{y}_i \geq 1 \text{ for all } i$$

- Recall regularization
- Optimization?

Dual formulation of SVM

- Constrained optimization

$$\min_w f(w) \text{ s.t. } g_i(w) \leq 0$$

- Consider the Lagrangian:

$$L(w, \alpha) = f(w) + \sum_i \alpha_i g_i(w)$$

- α_i 's: *dual variables*

- Claim: $\min_w f(w) \text{ s.t. } g_i(w) \leq 0$ is equivalent to:

$$\min_w \max_{\alpha \geq 0} L(w, \alpha)$$

- Min max interpretation: two player game, player trying to minimize picks w first, then player trying to maximize picks α second

Dual formulation of SVM

- Claim: $\min_w f(w)$ s. t. $g_i(w) \leq 0$ is equivalent to:

$$\min_w \max_{\alpha \geq 0} f(w) + \sum_i \alpha_i g_i(w)$$

- What happens if w doesn't satisfy the $g_i(w) \leq 0$ constraints?
 - What values will the α_i 's take?
- What happens if w satisfies all the $g_i(w) \leq 0$ constraints?
 - What values will the α_i 's take?
- This operation can be done for any constrained optimization problem

Weak duality

- Start with $\min_w f(w)$ s. t. $g_i(w) \leq 0$
- Rewrite it as $\min_w \max_{\alpha} L(w, \alpha)$, $L(w, \alpha) = f(w) + \sum_i \alpha_i g_i(w)$, $\alpha_i \geq 0$
- Claim (weak duality)

$$\max_{\alpha \geq 0} \min_w L(w, \alpha) \leq \min_w \max_{\alpha \geq 0} L(w, \alpha)$$

- Proof: For all $w, \alpha \geq 0$,

$$\min_{w'} L(w', \alpha) \leq L(w, \alpha)$$

Maximize left ineq: for all w , $\max_{\alpha \geq 0} \min_{w'} L(w, \alpha) \leq \max_{\alpha \geq 0} L(w, \alpha)$

Holds $\forall w$, minimize RHS over w : $\max_{\alpha \geq 0} \min_w L(w, \alpha) \leq \min_w \max_{\alpha \geq 0} L(w, \alpha)$

- But sometimes they're equal! *Strong duality*

Strong Duality

- Start with $\min_w f(w)$ s. t. $g_i(w) \leq 0$
- Rewrite it as $\min_w \max_{\alpha} L(w, \alpha)$, $L(w, \alpha) = f(w) + \sum_i \alpha_i g_i(w)$, $\alpha_i \geq 0$
- Claim (strong duality): Suppose f and g_i 's convex, and there exists a “strictly feasible” w such that $g_i(w) < 0$ for all i (Slater's condition). Then

$$\max_{\alpha \geq 0} \min_w L(w, \alpha) = \min_w \max_{\alpha \geq 0} L(w, \alpha)$$

- *Equal*, rather than $\leq$

Deriving dual formulation of SVM

Primal formulation of SVM:

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 \quad \text{s. t. } y_i(\langle w, x_i \rangle + b) \geq 1 \text{ for all } i$$

Rewrite constraints: $g_i(w, b) = 1 - y_i(\langle w, x_i \rangle + b) \leq 0$

Can rewrite into Lagrangian min-max form:

$$\min_{w,b} \max_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \frac{1}{2} \|w\|_2^2 + \sum_{i=1}^n \alpha_i (1 - y_i(\langle w, x_i \rangle + b))$$

Strong Duality

$$\min_{w,b} \max_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \frac{1}{2} \|w\|_2^2 + \sum_{i=1}^n \alpha_i (1 - y_i(\langle w, x_i \rangle + b))$$

Verify conditions for strong duality:

- $\frac{1}{2} \|w\|_2^2$ is convex
- $1 - y_i(\langle w, x_i \rangle + b)$ is affine, and hence convex
- A strictly feasible (w, b) exists. Why?
 - $1 < y_i(\langle w, x_i \rangle + b)$ for all i
 - Linearly separable dataset

Deriving dual formulation of SVM

Strong duality tells us that min-max and max-min probs have same val

$$\min_{w,b} \max_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \frac{1}{2} \|w\|_2^2 + \sum_{i=1}^n \alpha_i (1 - y_i (\langle w, x_i \rangle + b))$$

Becomes:

$$\max_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \min_{w,b} \frac{1}{2} \|w\|_2^2 + \sum_{i=1}^n \alpha_i (1 - y_i (\langle w, x_i \rangle + b))$$

Deriving dual formulation of SVM

$$\max_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \min_{w, b} \frac{1}{2} \|w\|_2^2 + \sum_{i=1}^n \alpha_i (1 - y_i (\langle w, x_i \rangle + b))$$

Fix some α for now, solve inner minimization. How? Set gradient = 0!

$$\frac{\partial}{\partial b} = - \sum_i \alpha_i y_i = 0, \quad \frac{\partial}{\partial w} = w - \sum_i \alpha_i y_i x_i = 0$$

Substitute into above and rearrange...

$$\min_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle - \sum_i \alpha_i \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0$$

Dual formulation of SVM. Only depends on x_i 's via dot products!!

Karush–Kuhn–Tucker (KKT) conditions

- For unconstrained convex f , $w^* = \arg \min_w f(w) \leftrightarrow \nabla_w f(w^*) = 0$
- What about for constrained optimization? KKT conditions
 - Characterize optimality assuming Slater's condition holds
- $\min_w f(w)$ s. t. $g_i(w) \leq 0$, let (w^*, α^*) be such that
 - Primal feasibility: $g_i(w^*) \leq 0$
 - Dual feasibility: $\alpha_i^* \geq 0$
 - Stationarity: $\nabla_w f(w^*) + \sum_i \alpha_i^* \nabla_w g_i(w^*) = 0$
 - **Complementary slackness**: for each $i \in [n]$, $\alpha_i^* g_i(w^*) = 0$
- Under convexity + Slater's condition, equivalent to: w^* is a minimizer of primal formulation, α^* is a maximizer of dual formulation.

Interpreting SVM solutions

$$\max_{\alpha \in \mathbf{R}^n, \alpha \geq 0} \min_{w, b} \frac{1}{2} \|w\|_2^2 - \sum_{i=1}^n \alpha_i (y_i (\langle w, x_i \rangle + b) - 1)$$

Complementary slackness:

$$\alpha_i (y_i (\langle w, x_i \rangle + b) - 1) = 0 \text{ for all } i \in [n]$$

Thus, either

$$\alpha_i = 0 \quad \text{or} \quad (y_i (\langle w, x_i \rangle + b) - 1) = 0$$

If $\alpha_i > 0$, these points are called *support vectors* (draw picture)

Note solution w can be written as combination of support vectors

$$w = \sum \alpha_i y_i x_i$$

Soft-Margin SVM

From Hard to Soft Margin SVMs

- (Draw non-separable, draw one outlier)
- Hard-margin:

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 \quad \text{s. t. } y_i(\langle w, x_i \rangle + b) \geq 1 \text{ for all } i$$

- Equivalently, if we let $\hat{y}_i = \langle w, x_i \rangle + b$

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 \quad \text{s. t. } 1 - y_i \hat{y}_i \leq 0 \text{ for all } i$$

- Soft-margin

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 + C \sum_i \max(0, 1 - y_i \hat{y}_i)$$

Interpreting Soft Margin

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 + C \sum_i \max(0, 1 - y_i \hat{y}_i)$$

- (draw cases of $y_i \hat{y}_i$, versus hard-margin case)
 - If $1 - y_i \hat{y}_i \leq 0$, zero hinge loss, on the correct side of margin
 - If $0 \leq y_i \hat{y}_i \leq 1$, correctly classified but within margin
 - If $y_i \hat{y}_i \leq 0$, incorrectly classified
- (draw hinge loss versus 0-1 loss, perceptron)
- If $C = 0$, ignore data, if $C = \infty$, hard-margin SVM
 - Larger C penalizes margin violations more

Deriving dual formulation of SVM

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 + C \sum_i \max(0, 1 - y_i \hat{y}_i)$$

- Define “slack variables” ξ_i
 - Interpretation: how far on wrong side of margin is point? (draw hinge loss)

$$\min_{w,b,\xi} \frac{1}{2} \|w\|_2^2 + C \sum_i \xi_i \text{ s. t. } \max(0, 1 - y_i \hat{y}_i) \leq \xi_i \text{ for all } i$$

- Break into two parts

$$\min_{w,b,\xi} \frac{1}{2} \|w\|_2^2 + C \sum_i \xi_i \text{ s. t. } 0 \leq \xi_i \text{ and } 1 - y_i \hat{y}_i \leq \xi_i \text{ for all } i$$

Deriving dual formulation of SVM

$$\min_{w,b,\gamma} \frac{1}{2} \|w\|_2^2 + C \sum_i \gamma_i \quad \text{s.t. } 0 \leq \xi_i \text{ and } 1 - y_i \hat{y}_i \leq \xi_i \text{ for all } i$$

- Introduce dual variables and take Lagrangian

$$\max_{\alpha, \beta \in \mathbb{R}^n, \alpha, \beta \geq 0} \min_{w,b,\xi} \frac{1}{2} \|w\|_2^2 + \sum_i (C \xi_i + \alpha_i (1 - y_i \hat{y}_i - \xi_i) - \beta_i \xi_i)$$

- Take derivative of inner problem, set to 0, substitute, simplify... (exercise)

$$\min_{\alpha \in \mathbb{R}^n, \alpha \geq 0} \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle - \sum_i \alpha_i \quad \text{s.t. } \sum_i \alpha_i y_i = 0$$

- Dual formulation of SVM. Only depends on x_i 's via dot products!!

Interpreting SVM solutions

$$\min_{\alpha \in \mathbf{R}^n, \textcolor{red}{C} \geq \alpha \geq 0} \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle - \sum_i \alpha_i \quad \text{s. t.} \quad \sum_i \alpha_i y_i = 0$$

- “complementary slackness” implies (after substitution $\beta_i = C - \alpha_i$)
 $\alpha_i(1 - y_i \hat{y}_i - \xi_i) = 0$ and $(C - \alpha_i)\xi_i = 0$
- If $\alpha_i = 0$, then $\xi_i = 0 \rightarrow y_i \hat{y}_i \geq 1$, point outside or on margin
- If $0 < \alpha_i < C$, then $\xi_i = 0$ and $y_i \hat{y}_i = 1$, point on margin
- If $\alpha_i = C$, then $y_i \hat{y}_i \leq 1$, if $\xi_i > 0$ then violates the margin

Optimization

- $\ell_{w,b}(x, y) = \max(0, 1 - y(\langle w, x \rangle + b))$
- Optimize loss function

$$L = C \sum_i \ell_{w,b}(x_i, y_i) + \frac{1}{2} \|w\|_2^2$$

- Note similarity to ridge regression (C on former vs λ on latter)
- Sub-gradient descent: $\frac{\partial L}{\partial w} = w + C \sum \delta_i$
 - $\delta_i = -y_i x_i$ if $1 - y_i \hat{y}_i \geq 0$, $\delta_i = 0$ if $1 - y_i \hat{y}_i \leq 0$ (draw, note non-diff pt)
- Could also run *projected* gradient descent on dual (draw, discuss)
- Solving dual using other methods are most practical